Clique identification in a Dirac-Hajós construction need not preserve criticality

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Abstract. Jensen and Toft ask whether a generalization of the Hajós construction, in which two k-critical graphs are identified along an h-clique before the usual edge deletion and addition, always produces a k-critical graph. We give a negative answer for every $k \geq 5$, already with $h = 3$. A 12-vertex 4-critical graph supplies the required coloring rigidity. For $k = 5$ the resulting graph has 23 vertices and 68 edges, and an edge e for which $\chi(D) = \chi(D - e) = 5$. The finite base-graph assertions are accompanied by complete coloring certificates and an exhaustive three-coloring table.

Keywords. Critical graph; chromatic number; Hajós construction; clique identification. **MSC 2020.** 05C15.

1. The question and result

A graph is k-critical if it is k-chromatic and every proper subgraph is $(k-1)$-colorable. In Problem 11.8 of Jensen and Toft [1], two disjoint k-critical graphs G_1, G_2 have distinguished h-cliques H_1, H_2, where $1 \leq h \leq k-2$. A vertex y_i outside H_i is adjacent to x_i in H_i. The cliques are identified so that x_1 and x_2 coincide; then $x_i y_i$ is deleted in each copy and $y_1 y_2$ is added. The question is whether the resulting graph must be k-critical. The case $h = 1$ is the usual Hajós construction; the question originated in Sachs and Stiebitz [2].

Theorem 1. For every integer $k \geq 5$ there are two k-critical graphs and an admissible choice $h = 3$ for which the construction produces a graph D_k containing an edge 01 such that $\chi(D_k) = \chi(D_k - 01) = k$. In particular, D_k is not k-critical. It has $2k + 13$ vertices and $k^2 + 15k - 32$ edges; D_5 has 23 vertices and 68 edges.

2. A finite four-critical base

Let B have vertex set $\{0, 1, \ldots, 11\}$. Table 1 specifies its edge set: $N^+(v)$ lists neighbors of v with larger labels. Thus $|E(B)| = 24$, $\{0, 1, 2\}$ is a triangle, and vertex 3 is adjacent to 0 but to neither 1 nor 2. We abbreviate an edge $\{u, v\}$ by uv.

v	$N^+(v)$	v	$N^+(v)$
0	1, 2, 3, 4, 11	6	9, 10
1	2, 6, 7, 10	7	8
2	9, 11	8	10
3	5, 8, 11	9	-
4	6, 7, 8, 9	10	-
5	6, 8	11	-

Table 1. Full edge specification of the base graph B.

Lemma 2. The graph B is 4-critical. Moreover, every proper 3-coloring c of B − 03 − 01 satisfies: if c(0) ≠ c(1), then c(3) = c(0); if c(0) = c(1), then c(3) = c(2).

Proof. Edges 02 and 12 remain. If c(0) ≠ c(1), relabel the colors so (c(0), c(1), c(2)) = (0,1,2); if equal, use (0,0,1). The complete list of three-colorings in these normalizations is:

colors on 0,1,2	complete color string
0, 1, 2	`012011202001`
0, 1, 2	`012022001121`
0, 0, 1	`001122110022`

The digit in position v, counting from zero, is c(v). To verify completeness, extend a coloring of 0,1,2 to vertices 3 through 11 in order, rejecting a color if an earlier adjacent vertex has it. The retained partial-coloring counts after assigning each vertex are:

last vertex	3	4	5	6	7	8	9	10	11
(0,1,2)	3	6	12	12	20	7	4	3	2
(0,0,1)	3	6	12	8	8	2	2	2	1

The first two strings are exactly the normalized 3-colorings of B − 03, since 01 is present there. Both give c(0) = c(3), so restoring 03 proves that B is not 3-colorable. Recolor vertex 3 with a fourth color in either string. Appendix A gives a proper 3-coloring for every edge deletion B − e. Every vertex has an incident edge; therefore B is 4-critical. □

3. Proof of Theorem 1

Put $r = k - 4 \geq 1$ and $G = K_r \vee B$, the join with an r-clique. Then G is k-critical. Indeed $\chi(G) = r + 4$. Deleting an edge within B permits r colors on the clique and three on B; deleting an edge within the clique permits r − 1 colors there and four on B. If an edge between a clique vertex q and v in B is deleted, color B − v with three colors, give v the color of q, and use r colors on the clique. Thus each edge deletion is (k − 1)-colorable.

Take two copies G_1, G_2 of G. Identify their copies of {0,1,2} labelwise, delete 03 in both copies, and add the edge $3^{(1)}3^{(2)}$. Call the result D_k. This is admissible because $h = 3 \leq k - 2$, with $x_i = 0$ and $y_i = 3^{(i)}$. Edge 01 remains.

Suppose $D_k - 01$ has a (k − 1)-coloring. Each r-clique uses r colors disjoint from its copy of B, so each B − 03 − 01 uses at most three colors. If the shared vertices 0 and 1 differ, Lemma 2 forces both $3^{(i)}$ to have the color of 0. If 0 and 1 agree, both $3^{(i)}$ have the color of 2. Either contradicts the edge $3^{(1)}3^{(2)}$. Therefore $\chi(D_k - 01) \geq k$.

Conversely, color B − 3 with colors 1,2,3 on both sides, agreeing on the shared triangle. Color $3^{(1)}$ with 4 and the first r-clique with 5,...,k. Color $3^{(2)}$ with 5 and the second r-clique with 4,6,...,k. This is a proper k-coloring of D_k. Hence $\chi(D_k) = \chi(D_k - 01) = k$.

Each input has $12 + r$ vertices and $24 + 12r + r(r - 1)/2$ edges. Identifying the triangles merges three vertices and counts their three edges once; the two deletions and one addition change the edge count by −1. Thus $|V(D_k)| = 2k + 13$ and $|E(D_k)| = k^2 + 15k - 32$. □

Reproducibility

The accompanying Python checker tests the finite certificates and sample outputs for k = 5,6,7. The Lean 4/mathlib file `D5Counterexample.lean` checks the 23-vertex output: edge 01 is present, the graph has a proper five-coloring, and deleting 01 still permits no four-coloring. The symbolic family is defined by `GeneralConstructionNative.lean`, and `GeneralNoKMinusOne.lean` proves that its output admits no coloring with $k - 1$ colors for every $k \geq 5$, using the base-coloring rigidity lemma. The general k-coloring and k-criticality of the input graphs have not yet been formalized in Lean. The equality linking the symbolic graph at $k = 5$ with the explicit edge list uses `native_decide`, which relies on Lean's compiler for that finite computation.

Code: https://github.com/alexgocardinal/dirac-hajos-jensen-toft-11-8.

Appendix A. Edge-deletion certificates for B

For every edge e, the twelve-digit string below colors $B - e$ properly with three colors. Its digit at position v colors vertex v. The ends of e have the same digit.

edge	coloring of B − e	edge	coloring of B − e
0-1	001122110022	3-8	012210202001
0-2	010112020222	3-11	012110202001
0-3	012011202001	4-6	012220201001
0-4	012200221101	4-7	012221020121
0-11	012110202000	4-8	012210201001
1-2	011112020222	4-9	012211020121
1-6	012220101001	5-6	012220001121
1-7	012221010121	5-8	012221001121
1-10	012211220011	6-9	012211020021
2-9	012211020221	6-10	012211220021
2-11	012110202002	7-8	012221000121
3-5	012222001121	8-10	012211220001

References

[1] T. R. Jensen and B. Toft, Graph Coloring Problems, Wiley, New York, 1995, §11.8, pp. 185–186. DOI: 10.1002/9781118032497.ch11.

[2] H. Sachs and M. Stiebitz, Colour-critical graphs with vertices of low valency, in Graph Theory in Memory of G. A. Dirac, Annals of Discrete Mathematics **41**, North-Holland, 1989, pp. 371–396.